WAKE FOREST UNIVERSITY SCHOOL OF MEDICINE
Curriculum Vitae

NAME Nathaniel S. O'Connell, PhD

ACADEMIC TITLE: Associate Professor

ADDRESS Department of Biostatistics and Data Science
Wake Forest University School of Medicine
Medical Center Boulevard
Winston-Salem, NC 27157
Office: (336) 713-1215
Website: https://nateoconnell.com/
Email: Nathaniel.oconnell@wfusm.edu

EDUCATION

2013	Kennesaw State University Kennesaw, Georgia Bachelors of Science (Biology)
2018	Department of Public Health Sciences Medical University of South Carolina Charleston, South Carolina PhD (Biostatistics) Research Advisor(s): Elizabeth Garrett-Mayer, PhD and Andrew Lawson, PhD Thesis: *Elicitation of Composite Patient Toxicity Scores in Cancer Clinical Trials with Application to Dose-Finding Trials for Combination Therapies*

EMPLOYMENT

Academic Appointments

Wake Forest University School of Medicine

2018 - 2026	Assistant Professor, Department of Biostatistics and Data Science
2020 - Present	Member, Wake Forest Comprehensive Cancer Center
2022 - Present	Member, Wake Forest Center for Artificial Intelligence Research
2022 – Present	Member, Wake Forest Center for Prevention Science in Child and Family Health
2024 – Present	Member, Wake Forest Center for Remote Health Monitoring
2025 – Present	Director of Biostatistics, Arise Cancer Consortium
2026 – Present	Associate Professor with Tenure, Department of Biostatistics and Data Science

Professional Experience

Medical University of South Carolina

2013 - 2018	Graduate Research Assistant, Department of Public Health Sciences

ADMINISTRATIVE SERVICE

Departmental Service

2020 – 2021	WFUSM Department of Biostatistics and Data Science Faculty Recruitment Committee
2020 – 2021	Member, COVID-19 Vaccine/Science Advocacy Group, Department of Biostatistics and Data Science
2025 – 2026	WFUSM Department of Biostatistics and Data Science Faculty Recruitment Committee

Institutional Service

2019 – Present	Protocol Review Committee Member, Comprehensive Cancer Center, Wake Forest University School of Medicine
2022	WFUSM Medical Student Research Program Application Reviewer
2022-2025	WFUSM Medical Student Research Day Poster Judge
2022 – 2023	WFUSM Data Science Interest Group, Planning and Steering Committee
2023 – 2024	WFUSM Research Advisory Committee (RAC)

Other Service and Memberships

2015 – 2016	Honor Council Member, College of Graduate Studies, Medical University of South Carolina
2016 – 2017	Admissions Committee Member, College of Graduate Studies, Medical University of South Carolina

EXTRAMURAL APPOINTMENTS AND SERVICE

Funding Agency Reviewer

American Society of Clinical Oncology: Conquer Cancer Foundation Grants Selection Committee, Biostatistical Reviewer, 2023-2026

NIH, Center for Scientific Review, Advancing Therapeutics A (ATA), [recurring] Biostatistical Reviewer, 2024 - Present (May 2024, October 2024, February 2025, June 2025, February 2026)

Department of Defense (DOD), FY24 Peer Reviewed Alzheimer's Research Program (PRARP) Transforming Care (TC), Biostatistical Reviewer, August 2024

Department of Defense (DOD), CDMRP's FY24 Traumatic Brain Injury and Psychological Health Research Program (TBIPHRP): Emerging Topics Research Award-Anger (ETR-A), Biostatistical Reviewer, December 2024

Department of Defense (DOD), CDMRP's FY24 Psychological Health Reseach Program -2, Biostatistical Reviewer, September 2025

Department of Defense (DOD), Alzheimers Research Program TR -4, Biostatistical Reviewer October 2025

Defense Health Agency (DHA), Multiple Sclerosis Research Program – Clinical Trials (MSRP-C T), Biostatistical Reviewer November 2026

Data Safety and Monitoring Board(s)

Medical University of South Carolina/OCI Biotech Limited, A Randomized Phase 1 Double Blind Placebo Controlled, Single-Dose and Multiple- Dose, Sentinel Lead-in, Dose-Escalation Study to Evaluate the Safety, Tolerability and Pharmacokinetics of Orally Inhaled Aerosolized OCI-B2 Suspension in Healthy Adult Volunteers up to 75 Years of Age, PI: Farhan Siddiqui, MD (Member, 2024 – Current)

Editorial boards

Childhood Obesity, Mary Ann Lieber Publishers inc.

*Nature Medicine – Neurology (*American Statistical Association (ASA) Statistical Advisory Panel).

Journal Reviewer

Clinical Trials (2019, 2021, 2022, 2023, 2025)
Statistics in Biosciences (2019)
Journal of the American Medical Association (2019)
Statistica Neerlandica (2020)
Statistics in Medicine (2020, 2026)
North Carolina Medical Journal (2021)
Neurology (2021, 2023)
Journal of Clinical Epidemiology (2022)
BMJ Open (2022, 2026)
Stat (2022)
Cancer (2022, 2024, 2025)
Communications in Statistics – Simulation and Computation (2023)
Expert Review of Respiratory Medicine (2023)
Nature Cancer (2023, 2024)
BMJ Public Health (2024)
Epilepsy and Behavior (2024)
Obesity Research and Clinical Practice (2025)
Nature Medicine (2025, 2026)
Journal of Clinical and Translational Science (2025)
Journal of Psychomatic Research (2025)
Cell Reports Medicine (2025)
Nature Communications (2025)
Trials (2026)
Journal of Science and Medicine in Sport (2026)

Conference Session Chair

Session Chair, 6th Early Phase Adaptive Trials Workshop, Nov 1-2 (2018), Cedars-Sinai Medical Center, Los Angeles, CA

Session Chair, Emerging Methods in Translational Science: Challenges and Opportunities in Aging and Cancer Research, Nov 9 (2018), Wake Forest School of Medicine, Winston-Salem, NC

Session Chair, [Virtual] Joint Statistical Meetings (JSM) 2021, Section on Statistical Learning and Data Science Speed Session D

Session Chair, Joint Statistical Meetings (JSM) 2022, Section on Statistical Learning and Data Science, Washington, D.C.

PROFESSIONAL MEMBERSHIPS AND SERVICE

2014 - Present American Statistical Association (ASA)

Biometrics Section (2016-Present)
Biopharmaceutical Section (2016-Present)
North Carolina Chapter (2018- Present)
Statistical Learning and Data Science Section (2020-Present)

2025-2026 ASA Statistical Learning and Data Science Conference 2026, Organizing Committee, Chair of Student Paper Award Review Committee.

2026-2027 North Carolina ASA Chapter – Vice President 2
(In-line for VP1 in 2027, Chapter President in 2028)

HONORS AND AWARDS

2009-2013	HOPE Scholarship, Kennesaw State University
2013	Graduate Recruitment Scholarship, Medical University of South Carolina
2015	Boyd Harshbarger Travel Award, Southern Regional Conference on Statistics
2015	Biopharm-Deming Student Scholar Award, Deming Conference and the American Statistical Association's Biopharmaceutical Section
2016	Outstanding Student Presentation Award, South Carolina American Statistical Association Palmetto Symposium
2016	Thomas C. Chalmers Award Finalist, Society for Clinical Trials, Annual Meeting
2016-2018	M. Clint Miller III Award, Department of Public Health Sciences, Medical University of South Carolina
2017	First Place Paper Award, American Statistical Association – Biopharmaceutical Section, Joint Statistical Meetings
2021	International Society of Quality of Life Research (ISOQOL) Annual Conference Young Investigator Award Finalist
2022	WFUSM Public Health Sciences Top 10 Paper Award (Publication #23)
2023	WFUSM Public Health Sciences Top 10 Paper Award (Publication #38)
2026	WFUSM Public Health Sciences 2020-2025 Most Influential Paper Award (Publications #23 and 38)

Career Development Programs (selective application)

2021-2022 Translational Research Academy Scholar, Clinical and Translational Science Institute, Wake Forest University School of Medicine

GRANT FUNDING

Active Funding

R01 HL159393 (PI: Alexander) 07/01/2022-06/30/2027 15%
Role: Site PI/Lead Biostatistician
Cardiovascular Impact of Near-complete Estrogen Deprivation for Breast Cancer
This multi-center clinical research study will quantify subclinical cardiovascular changes prospectively measured with cardiovascular magnetic resonance imaging over a two-year period in premenopausal women who undergo abrupt induction of menopause as part of their breast cancer therapy.

R01 AT011502 (PI: Wells) 07/01/2022-06/31/2027 10%
Role: Co-investigator/Lead Biostatistician
Online Techniques and Education Aimed to Manage Migraines
In this study, we are performing a prospective online clinical trial aimed at assessing the effects of headache education and mindfulness meditation at alleviation patient headache

burden. The primary outcome of interest is assessing reduction of migraine disability (measured by MIDAS) because of weekly practice of mindfulness-based stress reduction. Patients will be followed for 36 weeks.

1 R21 MD019434-01 (PI: Debinski) 10/2024 – 10/2026 10%
Role: Co-Investigator/Lead Biostatistician
Comparing Text Message and Community Health Worker Follow-Up for Missed Well Child Visits
This project conducts a randomized trial to compare well child visit attendance across different conditions to reduce disparities in visit schedule adherence, and determine the cost effectiveness for the different approaches. Specifically, we investigate the use of text message (low-touch) and community health worker outreach (CHW), compared to standard no standardized follow-up, in terms of improving rescheduling and completion of well-child visits in pediatric populations.

1UG3TR004803-01 (PI: Kirkendall) 08/2024-04/2029 20%
Role: Co-Investigator/Lead Biostatistician
Bytes to Bedside: Collaborative Development for Translational Clinical Decision Support
We propose to build a collaborative and innovative solution to the operational barriers to CDS evaluation through a Translational CDS Collaborative. The collaborative will identify effective CDS elements, advance CDS translational science and accelerate future translational research that uses CDS to implement research evidence into clinical care.

P30CA012197 **Wake Forest Baptist Comprehensive Cancer Center – Cancer Center Support Grant**
Role: Faculty Biostatistician 01/2022- 1/2027 5%
he Wake Forest Baptist Comprehensive Cancer Center supports the infrastructure for promoting transdisciplinary and translational research through the efforts of four scientific programs. These programs are supported by nine shared resources and a robust clinical research infrastructure as well as coordinated efforts in cancer research training and education and community outreach and engagement.

ARISE Cancer Consortium 01/2025 – Present 20%
Role: Director of Biostatistics
The Alliance for Research and Innovations in Pediatrics Oncology (ARISE) cancer consortium is philanthropically funded consortium, comprised of 10 sites in the Carolinas dedicated to conducting research studies and clinical trials in pediatric oncology. My role as the Director of Biostatistics is to oversee the development of study designs, data capture and monitoring, data analysis, and dissemination of results.

1R21CA300956-01 (PI: Vasbinder) 04/2025 – 03/2027 5%
Role: Co-investigator/Lead Biostatistician
Epigenetics, accelerated aging, and cardiovascular health during breast cancer treatment
This project investigates whether breast cancer treatment accelerates biological aging and how those epigenetic changes relate to declines in exercise capacity and cardiac function. Using longitudinal DNA methylation data and detailed cardiovascular phenotyping from the NCI-funded UPBEAT cohort, the study aims to identify mechanistic pathways linking treatment exposure, accelerated aging, and cardiovascular risk to inform future targeted interventions and improved risk stratification.

R01HL164434 (PI: Cilvik) 10/2024 – 8/2027 1%
Role: Co-Investigator/Lead Biostatistician
Developmental Origins of Cardiovascular Disease in Offspring from Non-Human Primate Pregnancies at Advanced Maternal Age
This project examines how advanced maternal age alters maternal cardiovascular adaptation, placental perfusion, and long-term cardiovascular health in offspring using a well-controlled non-human primate model. Leveraging longitudinal imaging, biomarker profiling, and adult

offspring phenotyping in the NIH-supported Vervet Research Colony, the study aims to identify mechanistic pathways of developmental programming that inform future interventions to reduce cardiovascular disease risk.

1 R01 AA 031477 – 01A1 (PI: Centanni) 08/2025 – 07/2029 5%
Role: Co-Investigator/Lead Biostatistician
Insula-BNST Circuit Regulation of AUD
Why do certain individuals progress from casual alcohol use to alcohol use disorder (AUD) while others do not? We aim to address this question by using preclinical models to examine stress and alcohol-induced adaptations in the insular cortex to bed nucleus of the stria terminalis network, a critical hub for AUD-related behavior. We will determine the role of this pathway in driving two key elements of AUD: compulsive-like drinking and hyperkatifeia during abstinence.

R260430 National Council of State Boards of Nursing (PI: Munn) 09/2026 08/2028 10%
Role: Co- Investigator/Biostatistician
Legislating Safety: A National Study of Workplace Violence Laws and Nurse Protection
The objective of this project is to evaluate variation in the comprehensiveness of U.S. state workplace violence (WPV) laws and examine whether more comprehensive legal approaches are associated with improved nurse safety and reduced severe occupational injury. Using legal epidemiology, a national nurse survey, and longitudinal injury surveillance data, the study assesses associations between state WPV law comprehensiveness, nurse-reported workplace violence experiences, and population-level injury trends. Findings are intended to inform nursing regulation and evidence-based legislative approaches to workforce protection.

Submitted Grants

R21 (PI: Jiang)
Role: Co-Investigator/Lead Biostatistician
Advancing Early-Stage NSCLC Recurrence Prediction through Foundation Model-Informed Deep Learning on Histopathology Images

R01 (PI: Jiang)
Role: Co-Investigator/Lead Biostatistician
Integrating Radiomics and Pathomics for Early Prediction of Peritoneal Metastasis in Gastric Cancer

R01 (PI: McGinnis, R)
Role: Co-Investigator/Lead biostatistician
Decision support for Assistive Devices in Multiple Sclerosis

Past Grant History

F31 CA210380, 8/1/2016 – 6/31/2018
NCI
Role: PI (Mentor: Elizabeth Garrett-Mayer)
Elicitation of Composite Patient Toxicity Scores in Cancer Clinical Trials with Application to Dose-Finding Trials for Combination Therapies
The purpose of this pre-doctoral grant is to develop novel methodology for eliciting continuous patient toxicity scores and incorporating them into drug-combination dose-finding trials in cancer trials.

1R01 CA199167 (PI: Hundley) 09/01/2015 - 12/31/2023
NHLBI $426,675
Role: Co-Investigator/Lead Biostatistician
UPBEAT : W. Understanding and Predicting Fatigue, CV Decline & Events After Breast CA Treatment
This multi-center study will fully characterize the time course of Adj-C cardiotoxicity and identify

factors that promote subclinical CV dysfunction, exercise intolerance, fatigue, and subsequent CV events among women of different ages and races/ethnicities, ultimately informing future therapies and risk factor prediction models to prevent CV events and thereby improve overall breast cancer- related survival.

R21 EB029493 (PI: Niazi) 04/1/2021 - 12/1/2023
NIH $193,750
Role: Co-Investigator
Development of Quantitative Tools to Predict Patients with Difficult Intubation to Minimize Treatment Related Complications
Role: Co-investigator
The overall goal is to develop a deep learning system to extract succinct features from profile views of the human head to identify patients with difficult intubation and develop an airway prediction model based on the synergy of frontal and profile features extracted through deep learning.

U01 CA233169 (Gray/Wagner) 09/18/2018 - 08/31/2024 2%
NCI $500,778
Analysis of ECOG-ACRIN Adverse Event Data to Optimize Strategies for the Longitudinal Assessment of Tolerability in the Context of Evolving Cancer Treatment Paradigms (EVOLV)
Role: Co-investigator
Improved methods for analyzing and interpreting adverse event data would lead to a more accurate and precise understanding of patients' experiences with cancer treatment, including newly emerging treatment regimens. The development of an effective approach to identifying patients who are experiencing (or who are at risk of experiencing) treatment tolerability issues could trigger improved AE management and maximize treatment outcomes.

UL1 TR001420, Pilot Grant (PI: O'Connell/Speiser) 07/01/2022-06/30/2023
5%
Role: Co-Principal investigator
Benchmarking random forest variable selection methods for continuous outcomes with open source data
This internal CTSI BERD pilot grant aims to evaluate random forest variable selection methods for continuous outcomes in terms of accuracy, computation time and simplicity using thousands of real-world, open source datasets.

R01 CA189824 (PI: Skelton) 10/01/2019 - 09/30/2024 10%
NCI $192,300
Role: Co-Investigator/Biostatistician/Lead Prediction Model Developer
Stay in Treatment: Predicting dropout from pediatric weight management study protocol
Role: Co-Investigator
The study will employ a model predictive of patient attrition that has been developed at Brenner FIT, the pediatric weight-management program directed by Skelton at Wake Forest Baptist's Brenner Children's Hospital. The researchers intend to test and refine the model's validity and accuracy through the collection and analysis of data from Brenner FIT and pediatric weight-management programs in Boston, Columbus, Ohio, and Kansas City, Mo.

R21 CA273665 (PI: Gurcan) 04/06/2023 – 03/31/2025 5%
Role: Co-Investigator/Lead Biostatistician
Efficient and cost-effective breast cancer risk stratification using whole slide histopathology images
We propose an automated method to predict Oncotype DX (ODX) Recurrence Score without annotations. If successful, this method would have a wide range of applications, including but not limited to the availability of an inexpensive, web-based tool to predict ODX in developing countries or rural areas with internet access.

BIBLIOGRAPHY

Peer-Reviewed Publications
*Denotes articles as first author (5), senior author/co-senior-author (8)

1. Golubski AJ, **O'Connell NS**, Schwartz JA, Ellermeyer SF. A 'crying wolf' game of interspecific kleptoparasitic mutualism. *Biology Letters*. 2014;10(3):20140073. doi:10.1098/rsbl.2014.0073 PMID: 24621556

2. Welch BM, **O'Connell NS,** Qanungo S, Halbert-Hughes C, Schiffman JD. Collecting Family Health History using an Online Social Network: a Nationwide Survey among Potential Users. *AMIA Annu Symp Proc*. 2015;2015:1316-1325. Published 2015 Nov 5. PMID: 26958272

3. Welch BM, **O'Connell N**, Schiffman JD. 10 years later: assessing the impact of public health efforts on the collection of family health history. *Am J Med Genet A*. 2015;167A(9):2026-2033. doi:10.1002/ajmg.a.37139 PMID: 25939339

4. Neitzke DJ, Bowers JS, Andrijauskaite K, **Oconnell, N.S.,** et al. Murine Th17 cells utilize IL-2 receptor gamma chain cytokines but are resistant to cytokine withdrawal-induced apoptosis. *Cancer Immunol Immunother*. 2017;66(6):737-751. doi:10.1007/s00262-017-1965-3 PMID: 28280853

5. ***O'Connell NS**, Dai L, Jiang Y, et al. Methods for Analysis of Pre-Post Data in Clinical Research: A Comparison of Five Common Methods. *J Biom Biostat*. 2017;8(1):1-8. doi:10.4172/2155-6180.1000334 PMID: 30555734

6. Aziz A, Pflieger L, **O'Connell N**, Schiffman J, Welch BM. Compatibility of Family History Cancer Guidelines With Meaningful Use Standards. *JCO Clin Cancer Inform*. 2017;1:CCI.17.00076. doi:10.1200/CCI.17.00076 PMID: 30148247

7. Welch BM, Harvey J, **O'Connell NS**, McElligott JT. Patient preferences for direct-to-consumer telemedicine services: a nationwide survey. *BMC Health Serv Res*. 2017;17(1):784. Published 2017 Nov 28. doi:10.1186/s12913-017-2744-8 PMID: 29183372

8. Ghoshal S, **O'Connell N**, Tegeler C, Freedman BI. Cerebral hemodynamics in peritoneal dialysis versus intermittent hemodialysis: A transcranial Doppler pilot study. *Perit Dial Int*. 2021;41(4):417-422. doi:10.1177/0896860820953712 PMID: 32909931

9. Wells RE, **O'Connell N**, Pierce CR, et al. Effectiveness of Mindfulness Meditation vs Headache Education for Adults With Migraine: A Randomized Clinical Trial. *JAMA Intern Med*. 2021;181(3):317-328. doi:10.1001/jamainternmed.2020.7090 PMID: 33315046

10. Reding KW, **O'Connell NS**, D'Agostino RB Jr, et al. Both intermuscular fat and LVEF decline promote heart failure symptoms in cancer survivors. *Cardiooncology*. 2021;7(1):16. Published 2021 May 8. doi:10.1186/s40959-021-00102-1 PMID: 33964981

11. Farris SR, Grazzi L, Holley M, Dorsett, A., Xing, K., Pierce, C.R., Estave, P., **O'Connell, N.**, Wells, R. Online Mindfulness May Target Psychological Distress and Mental Health during COVID-19. *Glob Adv Health Med*. 2021;10:21649561211002461. Published 2021 Mar 17. doi:10.1177/21649561211002461 PMID: 34497735

12. Semelka CT, Ornelles DA, **O'Connell NS**, et al. Detection of Environmental Spread of SARS-CoV-2 and Associated Patient Characteristics. *Open Forum Infect Dis*. 2021;8(6):ofab107. Published 2021 Mar 6. doi:10.1093/ofid/ofab107 PMID: 34183976

13. Estave PM, Beeghly S, Anderson R, Margol C, Shakir M, George G, Berger A, **O'Connell N**, Burch R, Haas N, Powers SW. Learning the full impact of migraine through patient voices: a qualitative study. *Headache*. 2021;61(7):1004-1020. doi:10.1111/head.14151 PMID: 34081779

14. Blackstock AW, Benson AB, Kudo M, Jimenez H, Achari PF, McGrath C, Kirchner V, Wagner LI, **O'Connell NS**, Walker K, Pasche VK. Safety and Efficacy of amplitude-modulated radiofrequency electromagnetic fields in advanced hepatocellular carcinoma. *4Open*. 2021;4:3. doi:10.1051/fopen/2021003 PMID: 41550248

15. Jordan JH, D'Agostino Jr RB, Ansley K, Douglas E, Melin S, Sorscher S, Vasu S, Park S, Kotak A, Romitti PA, **O'Connell NS**, Hundley WG, Thomas A. Myocardial function in premenopausal women treated with ovarian function suppression and an aromatase inhibitor. *JNCI Cancer Spectr*. 2021;5(4):pkab071. Published 2021 Jul 26. doi:10.1093/jncics/pkab071 PMID: 34476341

16. Meadows AD, Swanson SA, Galligan TM, Naidenko OV, **O'Connell N**, Perrone-Gray S, Leiba NS. Packaged Foods Labeled as Organic Have a More Healthful Profile Than Their Conventional Counterparts, According to Analysis of Products Sold in the U.S. in 2019-2020. *Nutrients*. 2021;13(9):3020. Published 2021 Aug 29. doi:10.3390/nu13093020 PMID: 34578899

17. Ip EH, Saldana S, Miller KD, Carlos RC, Gareen IF, Sparano JA, Graham N, Zhao F, Lee JW, **O'Connell NS**, Cella D. Tolerability of bevacizumab and chemotherapy in a phase 3 clinical trial with human epidermal growth factor receptor 2–negative breast cancer: A trajectory analysis of adverse events. *Cancer*. 2021;127(24):4546-4556. doi:10.1002/cncr.33992 PMID: 34726788

18. Huffman C, **O'Connell N**, Burns M, Blakely M, Gilliland W. Out and about: Factors associated with nurses' use of COVID-19 personal protective behaviors when not at work. *Nurs Forum*. 2022;57(2):234-243. doi:10.1111/nuf.12669 PMID: 34748227

19. Jones MK, **O'Connell NS**, Skelton JA, Halvorson EE. Patient characteristics associated with missed appointments in pediatric subspecialty clinics. *J Healthc Qual*. 2022;44(4):230-239. doi:10.1097/JHQ.0000000000000034 PMID: 35302524

20. *Raper JD, Thomas AM, Lupez K, Cox CA, Esener D, Boyd JS, Nomura JT, Davison J, Ockerse PM, Leech S, Johnson J, Abrams E, Murphy K, Kelly C, **O'Connell NS**, Weekes AJ. Can right ventricular assessments improve triaging of low risk pulmonary embolism?. *Acad Emerg Med*. 2022;29(7):835-850. doi:10.1111/acem.14484 PMID: 35289978 (co-senior author)

21. Obeng-Gyasi S, Graham N, Kumar S, Lee JW, Jacobus S, Weiss M, Cella D, Zhao F, Ip EH, **O'Connell N**, Hong F. Examining allostatic load, neighborhood socioeconomic status, symptom burden and mortality in multiple myeloma patients. *Blood Cancer J*. 2022;12(4):53. Published 2022 Apr 1. doi:10.1038/s41408-022-00648-y PMID: 35365604

22. Hughes RT, Tye KE, Ververs JD, **O'Connell NS**, Helis CA, Steber CR, Johnson AG, Chan MD, Farris MK. Virtual radiation oncology peer review is associated with decreased engagement and limited case discussion: Analysis of a prospective database before and during the COVID-19 pandemic. *Int J Radiat Oncol Biol Phys*. 2022;113(4):727-731. doi:10.1016/j.ijrobp.2022.04.026 PMID: 35489631

23. Hundley WG, D'Agostino R, Crotts T, Craver K, Hackney MH, Jordan JH, Ky B, Wagner LI, Herrington DM, Yeboah J, Reding KW, Ladd AC, Rapp SR, Russo S, **O'Connell NS**, Weaver KE, Dressler EV, Yaorong G, Melin S, Gudena V, Lesser GJ, Statins and left ventricular ejection fraction following doxorubicin treatment. *NEJM Evid*. 2022;1(9):10.1056/evidoa2200097. doi:10.1056/evidoa2200097 PMID: 36908314

*I was the primary biostatistician responsible for all statistical analyses conducted for and

presented in this paper – the primary paper from the PREVENT trial.

24. *Weekes AJ, Raper JD, Thomas AM, Lupez K, Cox CA, Esener D, Boyd JS, Nomura JT, Davison J, Ockerse PM, Leech S, Abrams E, Kelly C, **O'Connell NS**. Electrocardiographic findings associated with early clinical deterioration in acute pulmonary embolism. *Acad Emerg Med*. 2022;29(10):1185-1196. doi:10.1111/acem.14554 PMID: 35748352

25. Casacchia NJ, Rosenthal GE, **O'Connell NS**, Bundy R, Witek L, Wells BJ, Palakshappa D. Characteristics of adult primary care patients who use the patient portal: a cross-sectional analysis. *Appl Clin Inform*. 2022;13(5):1053-1062. doi:10.1055/a-1951-3153 PMID: 36167336

26. Gibbs KW, Forbes JL, **O'Connell NS**, Martin RS, Bakhru RN, Palakshappa JA, Files DC. Excess tidal volume ventilation in critically ill adults receiving adaptive pressure control: a cohort study. *Ann Am Thorac Soc*. 2022;19(11):1942-1945. doi:10.1513/AnnalsATS.202203-200RL PMID: 35622413

27. *Weekes AJ, Fraga DN, Belyshev V, Bost W, Gardner CA, **O'Connell NS**. Intermediate-risk pulmonary embolism: echocardiography predictors of clinical deterioration. *Crit Care*. 2022;26(1):160. Published 2022 Jun 4. doi:10.1186/s13054-022-04030-z PMID: 35659340

28. ***O'Connell NS**, Wages NA, Garrett-Mayer E. Quasi-partial order continual reassessment method: Applying toxicity scores to cancer dose-finding drug combination trials. *Contemp Clin Trials*. 2023;125:107050. doi:10.1016/j.cct.2022.107050 PMID: 36529437

29. Suddala S, **O'Connell N**, D'Agostino Jr R, Ladd A, Lucas AR, Jordan JH, Salloum FN, Ky B, Hackney MH, Hundley WG. Reduced Thoracic Aortic Distensibility Upon Breast Cancer Diagnosis (WF-97415 and WF-98213). *JACC CardioOncol*. 2023;5(5):704-706. Published 2023 Mar 7. doi:10.1016/j.jaccao.2022.11.018 PMID: 37969653

30. *Weekes AJ, Davison J, Lupez K, Raper JD, Thomas AM, Cox CA, Esener D, Boyd JS, Nomura JT, Murphy K, Ockerse PM, Leech S, Johnson J, Abrams E, Kelly C, **O'Connell NS**. Quality of life 1 month after acute pulmonary embolism in emergency department patients. *Acad Emerg Med*. 2023;30(8):819-831. doi:10.1111/acem.14692 PMID: 36786661

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32. Bellissimo MP, Reding KW, **O'Connell NS**, D'Agostino Jr RB, Mabudian L, Heiston EM, Ladd AC, Weaver KE, Lesser GJ, Fuemmeler BF, Salloum FN. Intermuscular Fat Changes Are Related to Reduced Left Ventricular Ejection Fraction During Cancer Therapy (PREVENT-WF-98213). *JACC CardioOncol*. 2023;5(3):409-411. Published 2023 Feb 20. doi:10.1016/j.jaccao.2023.01.006 PMID: 37397076

33. Lucas AR, D'Agostino Jr R, Jordan JH, Reding KW, **O'Connell NS**, Melendez GC, Salloum FN, Weaver KE, Lesser GJ, Hundley WG. Atorvastatin Does Not Attenuate Aortic Stiffening After Doxorubicin for Breast Cancer and Lymphoma (PREVENT-WF-98213). *JACC CardioOncol*. 2023;5(3):412-414. Published 2023 Jun 20. doi:10.1016/j.jaccao.2023.04.002 PMID: 37397089

34. *Weekes AJ, Raper JD, Esener D, Davison J, Boyd JS, Kelly C, Nomura JT, Thomas AM, Lupez K, Cox CA, Ockerse PM, Leech S, Johnson J, Abrams E, Murphy K, **O'Connell NS**. Comparing predictive performance of pulmonary embolism risk stratification tools for acute clinical deterioration. *Am Coll Emerg Physicians Open*. 2023;4(3):e12983. Published 2023 May 26. doi:10.1002/emp2.12983 PMID: 37251351

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72. Muthu N, Moses A, Orenstein E, Kandaswamy S, Chaparro JD, Molloy MJ, Hagedorn PA, Luo BT, McCoy AB, Fallon AA, Hess LM, Dziorny AC, Moore JB, Rigdon J, **O'Connell NS**, Kirkendall ES. Bytes to bedside: protocol for a multi-site mixed-methods study for development and evaluation of a collaborative framework for clinical decision support evaluation. *JMIR Preprints.* doi:10.2196/preprints.110523

73. Cochrane AC, McAllister V, Jensen ET, **O'Connell NS**. Patient- and provider-level factors associated with receipt of prescription medication for menopause-related symptoms. *Menopause.* In press 2026

Pre-prints of Submitted Papers (via arXiv or other official pre-prant channels)

1. Williams BM, Little JR, **O'Connell NS,** Centanni SW. A stress-activated mid-insula to BNST pathway regulates susceptibility to abstinence-induced negative affect in female mice. bioRxiv. 2025 Jan 7:2025-01.

2. **O'Connell NS**. Random forests as statistical procedures: design, variance, and dependence. arXiv preprint arXiv:2602.13104. Published 2026. https://arxiv.org/abs/2602.13104

3. **O'Connell NS,** Speiser JL. A Theory of Cross-Validation Variance, arXiv 2026

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Book Chapters

1. Garrett-Mayer E, **O'Connell N**. The evolution of phase I trials, past, present, and future: A biostatistical perspective. Novel Designs of Early Phase Trials for Cancer Therapeutics 2018 Jan 1 (pp. 17-32). Academic Press.

2. Chiuzan C, **O'Connell N**. Innovative phase I trials. Textbook of Clinical Trials in Oncology 2019 Apr 24 (pp. 27-51). Chapman and Hall/CRC.

Letters to the Editor/Responses

1. **O'Connell, N.S., "**Reply to S. Zhang et al. and S. Selukar" [Re: Importance of Low-and Moderate-Grade Adverse Events in Patients' Treatment Experience and Treatment Discontinuation: An Analysis of the E1912 Trial], *Journal of Clinical Oncology*, 2024

2. Wells, RE., Seminowics, D., **O'Connell, NS.,** Loder, E., Disagreement with Categorization of Clinical Trials included in Treadwell et al. Systematic Review and Meta-Analysis, *Headache*. 2025;65(9):1503-1504. doi:10.1111/head.15000 PMID: 40709627

Electronic Publications

1. Espeland M., **O'Connell N**., Roboussin, D., Rigdon, J., Shappel, H., Speiser JL, Decoding Vaccine Trials. https://school.wakehealth.edu/features/research/decodingvaccinetrials

2. **O'Connell, N**, Shappel, H., Espelan, M., Assuring COVID-19 vaccine safety. *Winston Salem Journal*, Jan 8 2021, https://journalnow.com/opinion/column/nathaniel-o-connell-heather-shappell-and-mark-espeland-assuring-covid-19-vaccine-safety/article_a7217bd2-5114-11eb-a463-e71e00b32bbb.html

Miscellaneous Publications

1. Garrett-Mayer E., **O'Connell N.S**. (2016, December 29) "Seamless phase II/III Trials" [video lecture], in The Biomedical & Life and Sciences Collection, Henry Stewart Talks, London, England. Online at https://hstalks.com/bs/3416/

PRESENTATIONS AT PROFESSIONAL MEETINGS

Regional

1. **O'Connell N.S**., Dai L., Jiang Y., Speiser J.L., Ward R., Wei W., Carroll R., Gebrehziabher M., "Bias and Efficiency Comparison of Methods for Analysis of Pre-Post Data." Poster presented at the Inaugural South Carolina Statistics Consortium hosted by the South Carolina Chapter of The American Statistical Association; 2014 Nov 15; Clemson University, Clemson, SC.

2. **O'Connell N.S**., Garrett-Mayer E., "A latent variable approach to elicit continuous toxicity scores and severity weights for multiple toxicities across varying grades in phase I oncology clinical trials." Poster presented at Southern Regional Conference on Statistics 2015 Summer Research Conference; 2015 June 7-10; UNC-Wilmington, Carolina Beach, NC.

3. **O'Connell N.S**., Garrett-Mayer E., "The Toxicity Score Elicitation Method: A Latent Variable Approach to Elicit Toxicity Scores for Multiple Toxicities in Oncology Clinical Trials" Presented talk at the 2016 South Carolina Chapter of the American Statistical Association Spring Meeting; 2016 March 25; University of South Carolina, Columbia, SC.

National

1. **O'Connell N.S**., Garrett-Mayer E., "A latent variable approach to elicit continuous toxicity scores and severity weights for multiple toxicities in dose-finding oncology trials." Poster presented at the 71st Deming Conference on Applied Statistics; 2015 December 7-11; Atlantic City, NJ.

2. **O'Connell N.S**., Garrett-Mayer E., "A latent variable approach to elicit continuous toxicity scores and severity weights for multiple toxicities in dose-finding oncology trials", oral presentation at Joint Statistical Meeting (JSM) - Baltimore; 2017 July 29- Aug 3; Baltimore, Maryland, USA.

3. **O'Connell N.S**., Garrett-Mayer E., "Elicitation of Continuous Toxicity Scores for use in Cancer Clinical Trials", poster presented at the 12th International Conference on Health Policy Statistics (ICHPS); 2018 Jan 10-12; Charleston, SC, USA.

4. **O'Connell N,** Estave P, Anderson R, Shakir M, Margol C, Beeghly S, Byrnes J, Coffield A, Seng E, Wells RE. Improvement in Migraine Disability from Mindfulness is Mediated by Improvements in Pain Catastrophizing and Depression. American Headache Society poster presentation 2021.

5. **O'Connell, N.S**., Skelton, J.A., "A prediction model method for optimizing appointment overbooking in healthcare clinics using electronic healthcare record data" oral presentation at Joint Statistical Meetings, August, 2021

6. **O'Connell N.S.,** et al., "Low and Moderate Grade Adverse Events Are Important Contributors to Patient-Reported Treatment Side-Effect Bother" oral presentation at ISOQOL, September 2021 **Selected as a New Investigator Poster Award Finalist*

7. **O'Connell NS,. "**Application of algorithmically defined toxicity index scores in dose-finding cancer trials." oral Presentation at Joint Statistical Meetings, August 2023

International

1. **O'Connell N.S**., Garrett-Mayer E., "A latent variable approach to elicit continuous toxicity scores and severity weights for multiple toxicities in cancer clinical trials." Oral presentation at the 37th Annual Meeting for the Society for Clinical Trials; 2016 May 15-18; Montreal, Quebec, Canada.

INVITED EXTRAMURAL PRESENTATIONS AND SEMINARS

Regional

1. **O'Connell, N.S.,** "A novel `dose-escalation` approach for determining an optimal methotrexate discharge level in pediatric osteosarcoma patients", Oral Presentation at ARISE Annual Meeting, 2025 April 24

National

1. **O'Connell NS,** Garrett-Mayer, E, "A latent variable approach to elicit continuous toxicity scores and severity weights for multiple toxicities in cancer clinical trials." Invited Talk at the 5th Early Phase Adaptive Trials Workshop; 2017 April; University of Virginia, Charlottesville, VA, USA

2. **O'Connell N.S**., Garrett-Mayer E., "Elicitation of Composite Patient Toxicity Scores in Dose-Finding Cancer Clinical Trials"; Invited Talk 2018 January 31; Mayo Clinic, Rochester, Minnesota.

3. **O'Connell N.S.**, Garrett-Mayer E., "The Partial Order Continual Reassessment Method for quasi-continuous toxicity scores" Invited Talk at the 6th Early Phase Adaptive Trials Workshop; 2018 November 1-2; Cedars-Sinai Medical Center, Los Angeles, CA, USA

4. **O'Connell N.S.,** "Cross Validation for Small and Imbalanced Data", Invited oral presentation within session on Practical Recommendations for Machine Learning that Advance Innovation, Joint Statistical Meeting (JSM) 2022 – Invited Session Oral Presentation, Washington, DC

5. **O'Connell N.S.,** Panel on Career Journey in the science of Cardiovascular Health, Invited Panel Presentation. Pauley Heart Center, Virginia Commonwealth University. 2024 January 17th, Richmond, VA

6. **O'Connell N.S.,** "Methods of Cross-Validation for Small and Imbalanced Data", Invited Talk, Department of Biomedical Informatics, The Ohio State University, 2024 August, 21

7. **O'Connell N.S.,** "Methods of Cross-Validation for Small and Imbalanced Data", Invited Talk Department of Biostatistics, University of Kentucky, 2024 September 18

8. **O'Connell N.S.,** "Methods of Cross-Validation for Small and Imbalanced Data", Invited Talk for the Section of Statistical Learning and Data Science (SLDS) of the American Statistical Association (ASA) monthly seminar series, 2024 November 19'

9. **O'Connell N.S.,** "The Importance of Low- and Moderate-Grade Adverse Events on Patient Side-Effect Burden in Cancer Trials and Example of their use in Dose-Finding Trial Methodology." The Mayo Clinic, Department of Quantitative Sciences - Grand Rounds, 2025 February 11th

International

1. **O'Connell N.S**., Garrett-Mayer E., "The Toxicity Score Elicitation Method: A Latent Variable Approach to Elicit Toxicity Scores for Multiple Toxicities in Oncology Clinical Trials" Invited talk at the 5th Early Phase Adaptive Trials Workshop; 2016 September 29-30; Politecnico Di Torino (Polytechnic University of Turin), Torino, Italy.

2. **O'Connell, NS,** "A Unified Framework for Cross-Validation: Theoretical Principled Guidance on how to perform CV for Prediction Performance", Invited Talk, International Chinese Statistical Association's Applied Statistics Symposium, June 2026

INSTITUTIONAL PRESENTATIONS

1. **O'Connell N.S**., Dai L., Jiang Y., Speiser J.L., Ward R., Wei W., Carroll R., Gebrehziabher M., "Bias and Efficiency Comparison of Methods for Analysis of Pre-Post Data." Poster presentation at Perry V. Halushka MUSC Student Research Day; 2014 Nov 14; Medical University of South Carolina, Charleston, SC.

2. **O'Connell N.S**., Garrett-Mayer E., "The Toxicity Score Elicitation Method: A Statistical Approach to Constructing Continuous Toxicity Scores for Multiple Toxicities in Phase-I Oncology Clinical Trials." Poster presentation at 15th Annual Holling's Cancer Center Research Retreat; 2015 Nov 13; Medical University of South Carolina, Charleston, SC.

3. **O'Connell N.S**., Garrett-Mayer E., "The Toxicity Score Elicitation Method: A Statistical Approach to Constructing Continuous Toxicity Scores for Multiple Toxicities in Phase-I Oncology Clinical Trials." Oral presentation at Perry V. Halushka MUSC Student Research Day; 2015 Nov 13; Medical University of South Carolina, Charleston, SC.

4. **O'Connell N.S**. "A Paradigm Shift in Cancer Dose-Finding/Phase I Trials and Methodology for Novel Endpoints" Departmental Brown Bag Seminar Series within the Department of Public Health Sciences; 2016 April 1; Medical University of South Carolina, Charleston, SC.

5. **O'Connell N.S**., Garrett-Mayer E., Lawson A., "Practical Elicitation and Implementation of Toxicity Scores to Represent Patient Toxicity Burden in Cancer Clinical Trials" Oral presentation at Perry V. Halushka MUSC Student Research Day; 2017 Nov 3; Medical University of South Carolina, Charleston, SC.

6. **O'Connell N.S**., Garrett-Mayer E., "The Quasi-Partial Order Continual Reassessment Method: Applying Toxicity Scores to Cancer Phase I Drug Combination Trials", oral presentation at Emerging Methods in Translational Science: Challenges and Opportunities in Aging and Cancer Research; 2018 Nov 9; Winston Salem, NC, USA.

7. **O'Connell N.S**., "Constructing and Implementing Continuous Toxicity Scores in Dose-Finding Cancer Trials", Ignite Talk presented at PHS 30th Anniversary Scientific Symposium; 2019

8. **O'Connell N.S.,** Halvorson, E., Skelton, J.A., "Optimization of "No Show" Prediction Models using EHR Data for Predictive Overbooking", poster presentation to Wake Forest Biomedical Informatics Colloquium, presented 2020 Oct 2; Winston Salem, NC, USA

9. **O'Connell N.S.,** "Cross Validation for Small and Imbalanced Data", Presented for Wake Forest Data Science Interest Group, 2022

10. **O'Connell N.S.,** "From Prediction to Inference: Quantifying Uncertainty in Model Evaluation", WFUSM Division of Public Health Science Grand Rounds, September 22, 2026

CONTRIBUTED PRESENTATIONS (As non-presenting author; only those with searchable abstracts on Google Scholar)

1. Wells R. E., Stewart A., Strauss L., Wise S., Granetzke L., Kumar S., Kumar V., Pierce C., Speiser J.L., **O'Connell N.S**., "Educating Patients on New Treatments: The Experience of One Headache Center's Launch of the New CGRP Migraine Medications" Oral presentation at 61st American Headache Society Annual Scientific Meeting; July 2019; Philadelphia, PA.

2. Ghoshal S., **O'Connell N.S**., Sarwal A., Tegeler C., Freedman B., "The impact on dialysis modality on cerebral hemodynamics: A transcranial Doppler study" Oral Presentation at Neurocritical Care Society Annual Meeting; 2019; Vancouver, Canada.

3. Stewart A., Young J., Strauss L., Wise S., Granetzke L., Kumar S., Kumar V., Pierce C., Sammons R., Speiser J., Guzik A., **O'Connell N.S**., Wells R. E., "Impact of multidisciplinary patient education sessions on new CGRP treatment expectations and understanding" Poster Presentation National Association of Specialty Pharmacy Annual meeting; 2019.

4. Figurelle, M.E., Sosa, C.H., **O'Connell, N.S**, Wells, R.E., Guzik, A.K., "*Increased Incidence of migraine in peri-menopausal aged women with stroke*" Poster Presentation American Academy of Neruology 2020 Meeting

5. Wells, R.E., **O'Connell, N.S.**, Pierce, C., Penzien, D., Loder, E., Houle, T., Zeidan, F., "*Mindfulness Meditation is More Effective at Reducing Pain Unpleasantness than Headache Education for Migraine: A Randomized Controlled Trial"* Oral Presentation IASP World Congress on Pain 2020

6. Zhao F, Peipert J, Lee JW, Hong F, Ip E, Gareen IF, **O'Connell N**, Carlos R, Mayer IA, Miller K, Partridge AH. Predictive value of bother by side effects of treatment prior to protocol therapy for early treatment discontinuation in clinical trials., Poster Presentation at American Society for Clinical Oncology (ASCO) annual meeting 2020

7. Peipert, J., Zhao, J., Lee, J., Hong, F., Ip, E., Gareen, I., **O'Connell, N.S**., Carlos, R., Stewart, K., Weiss, M., Sparano, J., Cella, D., Gray, R., Wagner, L., Increase in side effect bother was associated with early treatment discontinuation in a clinical trial among multiple myeloma patients. Oral Presentation American Society for Clinical Oncology (ASCO) annual meeting 2020

8. Wells, R. E., O'Connell, N., Pierce, C., Estave, P., Penzien, D., Loder, E., ... & Houle, T.,Mindfulness Meditation vs. Headache Education for Migraine: A Randomized Clinical Trial, Oral Presentation at American Academy of Neurology Meeting (4972), 2021

9. Sosa, C., Juneja, P., Kennedy, M., Figurelle, M., **O'Connell, N**., Wells, R., Guzik, A., Stroke Risk in preimenopausal women with migraine. Poster Presentation at International Stroke Conference 2021

10. Jordan, J., D'Agostino, R., **O'Connell, N**., Park, S., Kotak, A., Hundley, G., Thomas, A., Near Complete Estrogen Deprivation is Associated with Blunted Myocardial Perfusion Reactivity to Adenosine Stress in Pre-mopausal Hormone-Positive Breast Cancer Patients, Oral Presentation SCMR 2021

11. Peipert J, Zhao F, Lee JW, Hong F, Shen SE, Ip E, **O'Connell N**, Graham N, Smith ML, Gareen I, Carlos R. Change in a single-item indicator of treatment tolerability in cancer is associated with early treatment discontinuation. Oral Presentation In QUALITY OF LIFE RESEARCH 2021 Oct 1 (Vol. 30, No. SUPPL 1, pp. S51-S51).

12. Zhao F, Peipert J, Stearns V, Chang VT, Jegede O, Ip E, Lee JW, **O'Connell NS**, Graham N, Henry NL, Loprinzi CL. "Predictive value of baseline patient-rated treatment bother for early anastrozole discontinuation in a racially diverse cohort: Results from ECOG-ACRIN E1Z11"., Poster Presentation at American Society For Clinical Oncology (ASCO) Annual Meeting 2022

13. Shyamsunder, S., D'Agostino, R., **O'Connell, N. S**., Ladd, A., Weaver, K. E., Lesser, G. J., ... & Ge, Y.. Machine learning models for accurate pretreatment prediction of chemotherapy associated LV dysfunction in patients with breast cancer and lymphoma receiving chemotherapy (WF-98213 PREVENT and CCCWFU9912 DETECT IV). Poster Presentation at American Society For Clinical Oncology (ASCO) Annual Meeting 2022

14. Mabudian, L., **O'Connell, N. S**., D'Agostino, R. B., Heiston, E., Fleshman, J. A., Miller, D., ... & Hundley, W. G. "THE ASSOCIATION BETWEEN INTRAPERITONEAL FAT AND LEFT VENTRICULAR PERFORMANCE IN WOMEN RECEIVING CARDIOTOXIC CHEMOTHERAPY FOR STAGE I-IV BREAST, LYMPHOMA, OR SARCOMA." Poster Presentation - *Journal of the American College of Cardiology*, *79*(9_Supplement), 1828-1828., 2022

15. Suddala, S., **O'Connell, N. S**., D'Agostino Jr, R. B., Jordan, J. H., Meléndez, G. C., Hackney, M. H., ... & Hundley, W. G., PRIOR TO CANCER TREATMENT, DO WOMEN WITH BREAST CANCER EXPERIENCE REDUCED AORTIC DISTENSIBILITY IN THE ABSENCE OF HYPERTENSION?. Poster Presentation, *Journal of the American College of Cardiology*, *79*(9_Supplement), 1883-1883., 2022

16. Heiston EM, Mabudian L, **O'Connell NS**, D'Agostino R, Brubaker PH, Mihalko SL, Reding K, Bellissimo M, Ladd A, Lucas A, Jordan J. Impact of cardiometabolic health and abdominal adipose tissue on physical function in women with breast cancer (WF-97415). Abstract at American Society For Clinical Oncology (ASCO) Annual Meeting 2022

17. Reding K, **O'Connell NS**, Cheng R, Nguyen J, Ladd A, Jordan J, Ky B, Vasbinder A, Zaha V, Lucas A, Calhoun T. LVEF decline in relation to body composition among women treated for breast cancer: WF-97415. Abstract at American Society For Clinical Oncology (ASCO) Annual Meeting 2022

18. Reding K, **O'Connell NS**, Szewczyk W, Cheng RK, Nguyen J, Ladd AC, Jordan JH, Lucas AR, Sheppard VB, Ge Y, Salloum FN. Longitudinal Assessment of Racial Differences in Exercise Capacity After Breast Cancer Treatment: WF-97415. Abstract from AHA Scientific Sessions 2022, Circulation. 2022 Nov 8;146(Suppl_1):A15439-.

19. Sutton AL, **O'Connell NS**, Lucas AR, Franco RL, Reding KW, Kim Y, Cheng R, Sheppard VB, Weaver KE, Lesser GJ, Hundley WG. Examining the Association Between Socioeconomic Factors and Left Ventricular Ejection Fraction Declines in Breast Cancer and Lymphoma Survivors Receiving Doxorubicin: WF-98213. Abstract from AHA Scientific Sessions 2022. Circulation. 2022 Nov 8;146(Suppl_1):A12791-.

20. Douglas EH, Park CJ, **O'Connell NS**, Ansley KC, Richardson KM, Dent SF, Hackney MH, D'Agostino R, Hundley WG, Jordan JH, Thomas A. The CROWN study (CaRdiac outcomes with near complete estrogen deprivation): A multicenter, prospective cohort study of cardiovascular outcomes in premenopausal women with hormone receptor positive breast cancer treated with ovarian suppression and an aromatase inhibitor. Abstract Presentation at American Society for Clinical Oncology. 2023

21. Estave PM, Margol C, Beeghly S, Anderson R, Shakir M, Coffield A, Byrnes J, **O'Connell N**, Seng E, Gardiner P, Wells RE. Mechanisms of mindfulness in patients with migraine: Results of a qualitative study. Poster Presenttation. Headache: The Journal of Head and Face Pain. 2023 Mar;63(3):390-409.

22. Reding KW, Vasbinder A, **O'Connell N**, Demissei B, Szewczyk W, Cheng R, Ladd A, Lucas A, Meyer J, Bowen S, Salloum F. Biomarkers associated with atherosclerosis-related cardiovascular dysfunction in cancer survivors treated with anthracyclines. Poster Presentation. Cancer Research. 2024 Mar 22;84(6_Supplement):6296-.

23. Dent S, Douglas E, Hatcher S, Ansley K, Park C, **O'Connell N**, Hackney M, D'Agostino R, Bansal R, Avecilla R, Brown M. Abstract PO3-19-04: The CROWN STUDY: A multicenter prospective cohort study of cardiovascular outcomes in premenopausal women with hormone receptor positive breast cancer treated with ovarian suppression and an aromatase inhibitor. Poster Presentation. Cancer Research. 2024 May 2;84(9_Supplement):PO3-19.

24. Jabir AR, Ciarrocca AL, **O'Connell N**, Needles K, Pardee TS, Bhave R, Ellis LR, Wieduwilt MJ, Powell BL, Burkart M. The Impact of Weight Loss Grading Scale, a Measure of Cancer Cachexia, on Clinical Outcomes in Patients with Acute Myeloid Leukemia. Poster Presentation. Blood. 2024 Nov 5;144:2858.

25. Munn L, **O'Connell N**, Huffman C, Miller C, Stopyra J, Reed M, Gesell S. The Influence of Job-Related Factors on Nurse Well-Being. Poster Presentation, 2024 Annual Research Meeting 2024 Jul 1. AcademyHealth.

26. Lo, A., Hundley MD, G., Workman, S., Peral, R., Miller, D., **O'Connell, N.S.** and Ladd MD, A., 2025. 56-Body Composition Change in Women Treated for Breast Cancer before and 24 months after treatment. Presented as poster at Old Dominion University Student Symposium

27. Dantuluri, K.L., Liu, G., **O'Connell, N.S**., Tam Dellit, E.Q., Sanchez, P.J. and Ahmed, A., 2025, February. P-1126. Stratified Approach for Empiric Acyclovir in Neonates Presenting for Sepsis Evaluations. Poster presentation In *Open Forum Infectious Diseases* (Vol. 12, No. Supplement_1, pp. ofae631-1313). US: Oxford University Press.

28. Bellissimo, M., Ambrosio, M., Johnson, K., Canada, J., Jordan, J., Ladd, A., Brubaker, P., Mihalko, S., Dagostino, R., **O'connell, N.** and Ky, B., Physical Activity Preserves Exercise Capacity Over Two Years Of Breast Cancer Treatment (WF-97415). Poster Presentation at AHA 2025, *Circulation*, *152*(Suppl_3), pp.A4358258-A4358258.

29. Krishnan, V., Lambird, J., Pang, Y., **O'Connell, N.S**., Burkart, M., Powell, B.L., Howard, D. and Chojecki, A.L., Retrospective review of relapse after allogeneic stem cell transplant for myeloid malignancy. Poster Presentation ASCO 2025

30. Graham, M.E., **O'Connell, N**. and Cilvik, S., 2025, March. Advanced Maternal Age is Associated with Increased Offspring Birth Weight in a Nonhuman Primate Model. Poster Presentation. In *REPRODUCTIVE SCIENCES* (Vol. 32, No. 1, pp. 67A-67A). TIERGARTENSTRASSE 17, D-69121 HEIDELBERG, GERMANY: SPRINGER HEIDELBERG.

31. McAllister VM, **O'connell NS**, Jensen ET, Cochrane AC. Provider-Level Factors Related to Receiving Treatment for Menopause Symptoms. Poster Presentation. MENOPAUSE-THE JOURNAL OF THE MENOPAUSE SOCIETY 2025 Dec 1 (Vol. 32, No. 12). TWO COMMERCE SQ, 2001 MARKET ST, PHILADELPHIA, PA 19103 USA: LIPPINCOTT WILLIAMS & WILKINS.

32. Cochrane AC, Mcallister VM, Jensen ET, **O'connell NS**. Patient-level Factors Associated with Receipt of Prescription Medication for Menopause Symptoms. Poster. Presentation. MENOPAUSE-THE JOURNAL OF THE MENOPAUSE SOCIETY 2025 Dec 1 (Vol. 32, No. 12). TWO COMMERCE SQ, 2001 MARKET ST, PHILADELPHIA, PA 19103 USA: LIPPINCOTT WILLIAMS & WILKINS.

DIDACTIC/SYSTEMATIC INSTRUCTION

Medical University of South Carolina, Department of Public Health Sciences
Teaching Assistant, Foundations of Epidemiology I (3.0 hrs)
08/14 – 12/14

Medical University of South Carolina, Department of Public Health Sciences
Teaching Assistant, Introduction to Clinical Biostatistics (4.0 hrs)
08/15 – 12/15

Medical University of South Carolina, Department of Public Health Sciences
Teaching Assistant, Biostatistical Methods III – Correlated Data Analysis (3.0 hrs)
05/17 – 8/17

Wake Forest School of Medicine, Division of Public Health Sciences
Instructor, THSS 732: Statistical Modeling
01/21 – 5/21

Wake Forest School of Medicine, Division of Public Health Sciences
Instructor, THSS 732: Statistical Modeling
01/22 – 5/22

Wake Forest School of Medicine, Division of Public Health Sciences
Guest Lecturer, Power Calculations (MMTS 724)
Spring, 2021-2026

MENTORING RELATIONSHIPS

<u>Fellows</u>

2020-2022	Shivani Ghoshal, MD CTSI KL2 Research Scholar Mentoring Committee member
2020-2022	Ashish Khanna, MD CTSI KL2 Research Scholar Mentoring and THSS Capstone Committee member
2022	Charles Semelka, MD THSS Capstone Mentoring Committee
2021-2022	Nicholas Jarrod Casacchia, Pharm.D THSS Capstone Mentoring committee member
2022-2023	Halley Alexander, MD CTSI KL2 Research Scholar Mentoring and THSS Capstone Committee Member
2023-2024	Morgan Walls, MD CTSI KL2 Research Scholar Mentoring Committee Member
Summer 2024	Jerry Mensah Rawlings CAIR & CALIBER Summer Research Intern Mentor
Summer 2025	Olivia Harper CAIR & CALIBIR Summer Research Intern Mentor
2025-2029	Yashu Pokharel, MD NIH K-Scholar Statistical Mentor
2026-2028	Brad Rowland, MD CTSI KL2 Research Scholar Mentoring Committee
2026 -	James Adefisoye, PhD Assistant Professor Early Career mentoring Committee.

PUBLIC OUTREACH

2023	American Society of Clinical Oncology Role: Interviewee for ASCO News I was interviewed by ASCO Daily News to discuss my work related to publication #38 – The Importance of Low- and Moderate-grade Adverse Events on Treatment discontinuation and Patient Side-Effect Burden. The news article was posted online here: https://dailynews.ascopubs.org/do/low--and-moderate-grade-adverse-events-increase-odds-treatment-discontinuation, and would be featured on ASCO's monthly podcast in January 2024
2024	Oncology Times Role: Interviewee I was interviewed by Oncology Times regarding my work on low and moderate-

grade AEs from my published work in JCO (publication #38) in October 2023. My interview would be featured in a news article posted here: https://journals.lww.com/oncology-times/fulltext/2024/04000/3_questions_on_____low___moderate_grade_adverse.24.aspx